

ACHINTYA NIGAM

Cork, Ireland — +353 85 855 5886 — achintyanigam2@gmail.com
linkedin.com/in/achintyanigam — github.com/AchintyaNigam — [portfolio](#)

Education

Munster Technological University (MTU)

MSc Cybersecurity

Cork, Ireland

Sep 2026 – Present

Vellore Institute of Technology (VIT)

B.Tech Computer Science and Engineering — CGPA: 9.02/10

Vellore, India

2022 – 2026

Experience

Independ

Junior Software Developer

Vellore, India

Dec 2024 – Mar 2025

- Developed and maintained Kairos, a client management system, using React Native for Android.
- Implemented a background call-tracking service supporting automated call-data processing and secure handling of customer information.

Independ

Frontend Developer Intern

Vellore, India

Jul 2024 – Nov 2024

- Developed the Independ (formerly Xlancr) platform using Next.js.
- Collaborated with frontend and design teams to deliver the Xlancr landing page, contributing to improved customer retention.

Alumnus Software

Software Engineer Trainee Intern

Remote

Jan 2024 – Jun 2024

- Trained in developing full-stack web applications using Spring Boot and React.js, with emphasis on scalable application design.
- Developed full-stack applications using Spring Boot and React.js, implementing REST APIs, JWT authentication, and bcrypt password hashing.

Selected Projects

TCP/IP Network Traffic Analysis and Packet Crafting

Linux, tcpdump, Wireshark, hping3, Netcat

- Captured and analysed TCP/IP traffic using tcpdump and Wireshark, examining TCP sequence and acknowledgement numbers, connection establishment, data transmission, and packet behaviour.
- Constructed custom TCP packets using hping3 to investigate TCP flags, sequence numbers, acknowledgement numbers, window sizes, urgent pointers and IP fragmentation.

Post-Quantum Cryptography on Dual-Core Microcontrollers

C, RP2350A, Raspberry Pi Pico 2 W, Kyber, Dilithium

- Optimised CRYSTALS-Kyber and CRYSTALS-Dilithium cryptographic operations using the dual-core RP2350A architecture, achieving approximately 30% reduction in key-generation time for Kyber.
- Evaluated execution time, SRAM utilisation, power consumption and energy per operation to quantify the performance/energy trade-offs of multicore execution.
- Co-authored the resulting research, published in IEEE Access.

Technical Skills

Programming: Python, C, C++, Java, JavaScript, TypeScript

Cybersecurity: Network Security, TCP/IP, Vulnerability Assessment, Malware Analysis, SIEM, Security Monitoring

Security Tools: Wireshark, tcpdump, hping3, Nmap, Splunk

Web / Backend: React, Next.js, Spring Boot, REST APIs, HTML/CSS

Cloud / Databases: AWS, MySQL

Systems / Tools: Linux, Git

Certifications

AWS Certified Solutions Architect – Associate

2026

AWS Certified Cloud Practitioner

2024